

## **Price Prediction for Flights & 5\* Hotels from NY Airports to Cancun**

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**Project Description:** A machine learning predictive model to help maximize one's budget for a vacation in Cancun, utilizing trends to inform users on the best booking time for the best price and accommodations.

### **Problem Statement:**

The main issue that the creation of this project hopes to solve is the prediction of price fluctuations in real-time for booking flights to Cancun alongside a 5\* hotel resort. Cancun is a world-famous travel destination, with countless people booking reservations months or years ahead of time, and the issue becomes timing the booking correctly based on a wide variety of factors, including but not limited to: plane prices, weather conditions, tourist density, crime rate, hotel promotions, and hotel/excursion accessibility. We aim to create a model that aggregates information together from the various factors that affect booking prices to produce both a compelling product and project, as well as build a useful tool to incorporate into future planning when traveling with friends and family.

### **Strategic Aspects of Our Proposal:**

The initial strategy will be to clearly categorize all potential dataframes that could be associated with this project. As our project deals with real-world conditions that are always fluctuating, such as weather and price drops, it will be difficult to develop compact and efficient code that serves as consistent sorting, weighting, and cleaning algorithms. Once we organize all the necessary information into its proper dataframes, the prediction model developed to calculate price fluctuations will implement significant data management processes learned throughout the class as a form of machine learning. After categorizing our data, our project strategies will consist of:

**Intelligent Travel Decisions:** Through the utilization of data science and machine learning, price predictions determined on multiple factors and fluctuation trends will be able to be modeled. Some of the data aggregated and given varying weights in our price predictions include the number of days until departure, current flight price, historical flight price, hotel price trends, season/month, holiday/event indicators, and weather forecasts. Due to the nature of the data collected, an accurate optimization of elements between more consistent factors (holidays, seasons) and more unpredictable factors (weather) must be developed with a deep focus on data collection, rigorous cleaning, and model generation.

**Consumer Insights:** Based on our literature review and references for historical price ranges, accommodations for 5-star resorts in Cancun often lie in the same price range as “lower-star” stays. As such, centering our prediction modeling around 5-star hotels helps our users discover higher-end experiences still surprisingly in their budget range when compared to lower-rated hotels. To emphasize this focus on affordability, quality, reliability, and budget-indistinguishable factors, we decided to model the package to include all NYC and Newark flights. This way, regardless of the hotel that is being booked, there are reasonable, researchable, and high-quality accommodations factored into our predictive model, helping consumers finalize their travel plans with confidence.

**Predictive Modeling:** As the results from this project require real-time modeling for aiding price-efficient decisions, we must incorporate machine learning models involving regression, classification, and time-series forecasting into our predictive model. Regressions will utilize widely accessible knowledge and sales trends along designated flight routes and accommodations in our chosen destination (Cancun) to predict the future price of a flight or hotel. Classification machine models will use the data to predict whether the price will increase or decrease, providing a precise context to inform a user on whether to buy the plane ticket and book the hotel or wait until a different time or season. Time-series forecasting will capture patterns over time in price changes related to seasonal trends, severe weather events, and holiday spikes.

### **Significance of The Problem:**

International travel bookings are planned months in advance. With that in mind, how can travelers on a budget know the best time to book and what the weather conditions are during those months without extensive research? As a group of travelers who have spent hours creating one dubiously cost-effective itinerary after another, only for the price to drop further at some other time before or after we booked, the idea of a predictive model to better inform our decision-making was the perfect candidate. Combining personal experience and market trends with machine learning, a modeling system that indicates the timing of trips while accounting for conditions that can make or break an experience will offset the hours of research that go into traditional travel planning. Savvy users can derive insights that can save them hundreds of dollars and double their money’s worth in relief from the stresses of travel planning.

### **Issues:**

Traditional airline websites and hotel booking platforms mostly list current prices but fail to offer insights for when prices are likely to increase or decrease. They aren't incentivized to, as they would rather consumers buy travel packages when they are more expensive, for the sake of their own profit. Consequently, travelers often have to search manually and guess the best time to book their flights. By endeavoring to create a prediction model in this field, we are mechanizing the highly speculative world of travel pricing for a tool meant to serve consumers. Moreover, the hotel prices and flight prices can fluctuate due to seemingly unpredictable external factors such as hotel promotions, world events, and weather conditions. These fluctuations can be hard to detect, affect travel prices, and introduce noise to the prediction model while also affecting prediction accuracy. In addition to that, the past price data collected across various sources will vary widely and may include missing values, inconsistencies, messy formats, and duplicate entries that will require data cleaning and validation before analyzing and evaluating results.

### **Plan:**

We'll start by aggregating past flight/hotel price data from various sources, alongside secondary research or data on the influence of external factors such as holidays, seasonal trends, and world/weather events. Using historical price data as a basis the machine learning model can pull from, we're going to construct complex and multifaceted DataFrames with historical price data, including indicators from secondary influences, to help show the model what factors (weather, holidays, seasons) affect prices, and when during the year. After rigorously cleaning the data using Python and DataFrame operations, we will make the data we obtain queryable in PostgreSQL in order to streamline data processing into machine learning resources such as scikit-learn and Prophet. From there, we can visualize trends and influences on price with Matplotlib and Seaborn, which could help us determine potential weight calculations or multipliers for various observed factors, depending on how much a factor visually impacts historical price outcomes. With a cleaned and solidated PostgreSQL database, we will use Prophet's open-source forecasting library to help us build an accurate forecasting framework to analyze and interpret the results of our model's machine learning algorithms. All in all, we'll use Python, SQL for languages, PostgreSQL for the final database, scikit-learn and Prophet, a Meta-based open-source forecasting library that provides a framework for generating accurate forecasts. (from prophet import Prophet), for machine learning algorithms, Matplotlib and Seaborn for visualization tools, and Prophet for predictive forecasting.